

Answer Key & Teacher Notes

Session 1 – Dividing a Fraction by a Fraction • Session 2 – Dividing with Mixed Numbers

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Dividing a Fraction by a Fraction

TALK PROMPTS

- Do you need a common denominator to divide two fractions? Explain your thinking.
- Why does multiplying by the reciprocal give the same answer as dividing?
- Neither number is a whole pound. Why is $7/8 \div 1/16$ still a whole number?

WATCH FOR

- Flipping both fractions, or flipping the dividend instead of the divisor ($3/5 \div 1/2$ written as $5/3 \times 2/1$).
- Multiplying straight across without flipping ($5/12 \div 5/12$ written as $25/144$).
- Expecting a quotient less than 1 because both numbers are less than 1.

PARTNER TASK

Honey Jars. One partner writes $3/4 \div 1/16$ for $3/4$ pound of honey and $1/16$ pound per jar; the other rewrites it as multiplication and solves (12 jars). Switch roles with Sanding Strips: $11/12$ meter torn into $1/12$ -meter strips (11), and draw the tape diagram.

SESSION 2 • TEACHER NOTES

Dividing with Mixed Numbers

TALK PROMPTS

- Why can you not flip a mixed number the same way you flip a fraction?
- After converting, $3 \frac{3}{4}$ and $1 \frac{1}{4}$ had the same denominator. Did that make the division easier? Say why.

WATCH FOR

- Converting a mixed number by adding the whole and numerator ($4 \frac{1}{4}$ written as $5/4$).
- Flipping a mixed number piece by piece ($1 \frac{1}{4}$ flipped to $1 \frac{4}{1}$) instead of converting first.
- Dividing whole parts and fraction parts separately ($8 \frac{3}{4} \div 1 \frac{1}{4}$ as $8 \div 1$ and $3/4 \div 1/4$).

PARTNER TASK

Paint Coats and Yarn Scarves. One partner converts $3 \frac{3}{4}$ and $1 \frac{1}{4}$ to improper fractions; the other divides (3 coats). Switch roles for $11 \frac{1}{4} \div 2 \frac{1}{4}$ (5 scarves), then check each answer by multiplying back.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.NS.A.1	1	2	1	1	2	2	3	2	3	1	1
S1 • Form B	6.NS.A.1	1	2	1	1	2	2	3	2	3	1	1
S2 • Form A	6.NS.A.1	1	2	1	1	2	2	3	2	3	1	1
S2 • Form B	6.NS.A.1	1	2	1	1	2	2	3	2	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

reciprocal: A fraction flipped upside down; the two numbers multiply to make **1**.

dividend: The amount you **start** with and split into groups.

divisor: The size of each group; the number you divide **by**.

- To divide by a fraction, multiply by its **reciprocal**.
- Keep the first fraction, change $\div$ to $\times$, and flip only the **second** fraction.
- Check: quotient $\times$ divisor = **dividend**.

Example: Write the division: $3/4 \div 1/8$. Multiply by the reciprocal: $3/4 \times 8 = 24/4$. Simplify: $24/4 = 6$. Answer: **6** strips. Check: $6 \times 1/8 = 6/8 = 3/4$.

YOU TRY

- a. 3 b. $1/2$
- $9/10 \div 3/10 = 9/10 \times 10/3 = 3$ bags

EXIT TICKET

- a. 4 b. 6
- $7/8 \div 1/8 = 7/8 \times 8 = 7$ muffins
- A** ($3/5 \times 7/2$)

PRACTICE FORM A

- a. $5/2 = 2\ 1/2$ b. $3/2 = 1\ 1/2$ c. 4 d. 6 e. $2/5$ f. $5/6$
- a. $3/4 \div 1/12 = 3/4 \times 12 = 36/4 = 9$ bowls b. $4/5 \div 2/5 = 4/5 \times 5/2 = 20/10 = 2$ bags
- $11/12 \div 1/12 = 11/12 \times 12 = 11$ strips
- $7/8 \div 1/16 = 7/8 \times 16 = 112/8 = 14$ jars
- Nadia is wrong: she multiplied $5/12 \times 5/12$ without flipping. $5/12 \div 5/12 = 5/12 \times 12/5 = 1$ piece. A number divided by itself is 1.
- a. $15/16 - 3/16 = 12/16 = 3/4$ L b. $3/4 \div 1/16 = 3/4 \times 16 = 12$ jars
- a. $9/10 \div 3/10 = 3$ large projects b. $9/10 \div 1/10 = 9$ small projects c. $3/10$ is 3 times $1/10$, so 3 times as many small pieces fit in the same $9/10$ meter.
- a. $5/6 \div 1/18 = 5/6 \times 18 = 90/6 = 15$ name tags b. $1/18$ is a very small piece, so many of them fit in $5/6$ yard.
- Only the divisor flips. Keep $3/5$, change $\div$ to $\times$, flip $1/2$ to $2/1$: $3/5 \times 2/1 = 6/5 = 1\ 1/5$.
- B** (3)
- C** (reciprocal)

Self-check word DIVISOR

PRACTICE FORM B

- a. $7/2 = 3\ 1/2$ b. $6/5 = 1\ 1/5$ c. 5 d. 8 e. $3/8$ f. $5/3 = 1\ 2/3$
- a. $5/6 \div 1/18 = 5/6 \times 18 = 90/6 = 15$ bottles b. $6/7 \div 2/7 = 6/7 \times 7/2 = 42/14 = 3$ portions
- $9/10 \div 1/10 = 9/10 \times 10 = 9$ strips
- $3/4 \div 1/16 = 3/4 \times 16 = 48/4 = 12$ jars
- Kofi is wrong: he multiplied $3/8 \times 3/8$ without flipping. $3/8 \div 3/8 = 3/8 \times 8/3 = 1$ piece. A number divided by itself is 1.
- a. $11/12 - 3/12 = 8/12 = 2/3$ L b. $2/3 \div 1/24 = 2/3 \times 24 = 16$ jars
- a. $4/5 \div 2/5 = 2$ large banners b. $4/5 \div 1/10 = 8$ small banners c. $2/5 = 4/10$, which is 4 times $1/10$, so 4 times as many small pieces fit in the same $4/5$ meter.
- a. $2/3 \div 1/15 = 2/3 \times 15 = 30/3 = 10$ name tags b. $1/15$ is a very small piece, so many of them fit in $2/3$ yard.
- Only the divisor flips. Keep $4/7$, change $\div$ to $\times$, flip $2/3$ to $3/2$: $4/7 \times 3/2 = 12/14 = 6/7$.
- B** (4)
- C** ($7/3$)

Self-check word INVERSE

GUIDED NOTES

mixed number: A **whole** number and a fraction written together.

improper fraction: A fraction whose numerator is **greater** than or equal to its denominator.

simplify: Write the same amount using **smaller** numbers.

- Before dividing, write each mixed number as an **improper** fraction.
- Then multiply by the **reciprocal** of the divisor and simplify.
- You cannot flip a mixed number until it is **one** fraction.

Example: Convert: $3\frac{1}{2} = \frac{7}{2}$ and $1\frac{3}{4} = \frac{7}{4}$. Multiply by the reciprocal: $\frac{7}{2} \times \frac{4}{7} = \frac{28}{14}$. Simplify: $\frac{28}{14} = 2$. Answer: She can make **2** loaves.

YOU TRY

1. a. $\frac{5}{2}$ b. $10\frac{3}{3} \div \frac{3}{2} = 5$
2. $15\frac{4}{4} \div \frac{3}{4} = 15\frac{4}{4} \times \frac{4}{3} = 5$ servings

EXIT TICKET

1. a. $\frac{13}{4}$ b. 5
2. $15\frac{2}{2} \div \frac{3}{2} = 15\frac{2}{2} \times \frac{2}{3} = 5$ pieces
3. **A** ($\frac{7}{3} \times \frac{6}{7}$)

PRACTICE FORM A

1. a. $\frac{22}{5} = 4\frac{2}{5}$ b. $\frac{9}{2} = 4\frac{1}{2}$ c. 3 d. $\frac{15}{2} = 7\frac{1}{2}$ e. $\frac{1}{2}$ f. $\frac{2}{3}$
2. a. $4\frac{1}{2} = \frac{9}{2}$; $\frac{9}{2} \div \frac{3}{4} = \frac{9}{2} \times \frac{4}{3} = \frac{36}{6} = 6$ pieces b. $15\frac{2}{2} \div \frac{5}{2} = 15\frac{2}{2} \times \frac{2}{5} = 3$ bags
3. $15\frac{4}{4} \div \frac{5}{4} = 15\frac{4}{4} \times \frac{4}{5} = 3$ coats
4. $45\frac{4}{4} \div \frac{9}{4} = 45\frac{4}{4} \times \frac{4}{9} = 5$ scarves
5. Tomás is wrong: he found $4\frac{1}{2} \times \frac{1}{2}$. $4\frac{1}{2} \div \frac{1}{2} = \frac{9}{2} \times 2 = 9$ scoops. Each cup holds 2 half-cup scoops, so there must be more scoops than cups.
6. a. $16\frac{1}{2} - 4\frac{1}{2} = 12$ ft b. $12 \div 1\frac{1}{2} = 12 \times \frac{2}{3} = 8$ pieces
7. a. $7\frac{1}{2} \div 1\frac{1}{4} = \frac{15}{2} \times \frac{4}{5} = 6$ spacings b. 6 markers c. No marker sits at 0, so each spacing ends at exactly one marker.
8. $\frac{28}{3} \div \frac{7}{3} = \frac{28}{3} \times \frac{3}{7} = 4$ pitchers. Check: $4 \times 2\frac{1}{3} = 9\frac{1}{3}$.
9. $4\frac{1}{4}$ is $4 \times 4 + 1 = 17$ fourths, not $\frac{5}{4}$ (he added 4 + 1). Correct: $\frac{17}{4} \times \frac{2}{1} = \frac{34}{4} = 8\frac{1}{2}$.
10. **A** ($\frac{11}{4}$)
11. **B** (make improper fractions)

Self-check word FLIPPED

PRACTICE FORM B

1. a. 4 b. 3 c. 5 d. $\frac{40}{3} = 13\frac{1}{3}$ e. $\frac{1}{2}$ f. $\frac{3}{5}$
2. a. $5\frac{1}{4} = \frac{21}{4}$; $\frac{21}{4} \div \frac{3}{8} = \frac{21}{4} \times \frac{8}{3} = \frac{168}{12} = 14$ sections b. $\frac{35}{4} \div \frac{7}{4} = \frac{35}{4} \times \frac{4}{7} = 5$ tubs
3. $\frac{20}{3} \div \frac{5}{3} = \frac{20}{3} \times \frac{3}{5} = 4$ coats
4. $\frac{27}{2} \div \frac{9}{4} = \frac{27}{2} \times \frac{4}{9} = \frac{108}{18} = 6$ scarves
5. Elena is wrong: she found $5\frac{1}{2} \times \frac{1}{2}$. $5\frac{1}{2} \div \frac{1}{2} = \frac{11}{2} \times 2 = 11$ scoops. Each cup holds 2 half-cup scoops, so there must be more scoops than cups.
6. a. $14\frac{1}{4} - 4\frac{1}{4} = 10$ ft b. $10 \div 1\frac{1}{4} = 10 \times \frac{4}{5} = 8$ pieces
7. a. $10 \div 2\frac{1}{2} = 10 \times \frac{2}{5} = 4$ spacings b. 4 stations c. No station sits at 0, so each spacing ends at exactly one station.
8. $\frac{35}{4} \div \frac{5}{4} = \frac{35}{4} \times \frac{4}{5} = 7$ pitchers. Check: $7 \times 1\frac{1}{4} = 8\frac{3}{4}$.
9. $3\frac{2}{5}$ is $3 \times 5 + 2 = 17$ fifths, not $\frac{5}{5}$ (she added 3 + 2). Correct: $\frac{17}{5} \times \frac{5}{4} = \frac{17}{4} = 4\frac{1}{4}$.
10. **B** ($\frac{14}{3}$)
11. **C** (2)

Self-check word INVERTS

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 · DIVIDING A FRACTION BY A FRACTION

1. $4/5 \div 1/10 = 8$
2. a. 6 b. 3 c. $5/2 = 2 \frac{1}{2}$ d. $7/4 = 1 \frac{3}{4}$ e. $2/3$ f. 2
3. $1/2 \div 2/3 = 1/2 \times 3/2 = 3/4$ of the recipe
4. $5/6 \div 5/12 = 5/6 \times 12/5 = 60/30 = 2$ bookmarks
5. She did not flip the divisor (she multiplied by $2/3$ instead of its reciprocal $3/2$). Correct: $4/5 \times 3/2 = 12/10 = 6/5 = 1 \frac{1}{5}$.

SHEET 2 · DIVIDING WITH MIXED NUMBERS

1. a. 6 b. 4 c. $1 \frac{1}{2}$ d. $4 \frac{1}{2}$ e. 5 f. $3/4$
2. a. $17/5$ b. $29/6$
3. $8 \frac{3}{4} \div 1 \frac{1}{2} = 35/4 \times 2/3 = 35/6 = 5 \frac{5}{6}$, so 5 full bowls; $5 \times 1 \frac{1}{2} = 7 \frac{1}{2}$, and $8 \frac{3}{4} - 7 \frac{1}{2} = 1 \frac{1}{4}$ cups left
4. $7 \frac{1}{2} \div 1 \frac{1}{4} = 15/2 \times 4/5 = 60/10 = 6$ hours
5. less than $6 \frac{1}{2}$ (the divisor is greater than 1; $6 \frac{1}{2} \div 1 \frac{1}{4} = 5 \frac{1}{5}$)
6. **B** (2)
7. **C** (8)